Actuarial Pricing Across Markets with AI Assistance — Solo entry (Suguru Fujita)

Final version (v1.0, submitted 2026-10-30). The Week 2 private score is published after the close. Every number below is traceable to `logs/` *or* `outputs/reports/`*. Graded scope: first 5 pages.*

1. Executive summary

- **Task.** Expected 2025 ultimate claim cost (risk premium) per policy row for private motor, home buildings and home contents; metric Tweedie D^2 (p = 1.5). Week 1: developed market. Week 2: an emerging market with two years of history, whose 2025 test book is mostly new business from a broker.
- **Approach.** Target rebuilt from claims and reconciled to the cent; development to ultimate (cost-based IBNR, Bornhuetter-Ferguson override for immature bodily injury); mix-adjusted trend; per-line 50/50 blend of a Tweedie GLM (explainability anchor) and a regularised LightGBM Tweedie, with large-loss caps + loading and a credibility-weighted experience (bonus-malus) modifier. In Week 2 the rating shape was carried over from the developed market and the level re-fitted on emerging data, with a new-business loading.
- **Results.** Week 1 temporal validation (train 2021–23 → validate 2024, pooled): constant 0.082 → GLM 0.137 → blend 0.139; **private D^2 0.13363, rank 30** (public 0.12480). The private score was close to our temporal-validation estimate, consistent with selecting models on validation rather than on the leaderboard. Week 2 validation (train 2023 → validate 2024 on emerging): motor 0.082, buildings 0.102, contents 0.102; public 0.15691 (private score published after the close).
- **Office premium.** For the 20 emerging reference policies, the office premium built from our risk premium is a median 2.7× the current written premium, mostly because of the emerging expense structure (§5).
- **AI workflow.** Claude Code built and challenged every step; the human owner signed off 38 logged decisions at six review gates. Several AI errors were caught by tests, validation and the official data description before they reached a submission (§3).

2. Pricing performance & adaptation

2.1 Week 1 — developed market

Data and target (gate G1). Target per claim = max(incurred − recoveries, 0); amounts are already net of excess (official description, confirmed on data: 1% quantile of closed claims ≈ 13–41 CU at every excess level). Declined claims (1,711) and claims outside any row period (414; 0.24% of cost) excluded. All 163,150 claims covered; claim-level vs row-level reconciliation difference 0.00 in all 12 line × year cells.

Data quality (gate G2). Train exposure recomputed from dates (14,204 rows had days or ~1.4% inflation; raw exposure would understate burning cost by ~26%); sum insured ×1,000 entry errors corrected with the train median rule (train and test, ~0.2%); construction type relabel T (2021–22) → W (2023–24) unified as timber, with a timber risk shift flagged. One open bodily-injury claim at 11.89M (next largest 1.12M) treated as a genuine large loss with a sensitivity at 1.19M.

Development and trend (gate G3). Single valuation date (31 Dec 2024), so incurred development (IBNER) is not estimable; cost-based reporting-delay factors per line × peril (2024: motor 1.32, buildings 1.12, contents 1.04, central). Bodily-injury severity for 2023–24 falls as the open share rises (49%/80%), so it is replaced by the 2021–22 severity trended (+3% p.a.). Trend from mix-adjusted GLM year effects and per-peril analysis:

Annual trend	Low	Central	High
Motor	3%	5%	7%
Buildings	4%	5%	7%
Contents	2.5%	3.5%	4.5%

Insight: motor's crude frequency fall is portfolio mix (mix-adjusted frequency flat); physical-damage severity +6–10% p.a. Leaderboard probes of low/high trend were within noise of central, suggesting level is not the main source of error.

Model ladder (temporal validation, 2024 fold).

Rung	Pooled D^2	Pooled A/E
Constant per line	0.0820	0.98
Tweedie GLM (capped + loading)	0.1370	0.97
GLM + LightGBM blend	0.1388	0.99

Rolling origin (→2022, →2023, →2024): the blend beats the GLM in 8 of 9 line × year folds. Level sensitivity: the D^2-optimal global scale is 0.96–1.03 (gain ≤ 0.0002), suggesting the remaining error lies mostly in ranking.

Adopted vs rejected (pre-registered rule: +0.001 D^2 in BOTH 2023 and 2024 folds). Adopted: caps 100k/100k/25k; contents GBM c7 (+0.0018/+0.0017). Rejected: recency weighting (motor −0.0037 in 2023), written premium as a feature, raw make_model, prior-claims features v1/v2, stacking (unstable weights). Leaderboard-informed (documented as such): long-run instead of same-year weather (+0.0011 LB; also the principled choice — realised weather is unknown at pricing); near-pass buildings c7 and experience modifier (b3, public +0.0004, the two effects adding up; a pre-registered re-test on the final model confirmed the modifier as a near-pass: +0.0002 to +0.0011 in all six cells, only motor 2024 ≥ 0.001). Final selection: validated model (b) plus b3 (Kaggle scores the better of the two); step-by-step scores in Appendix A.

Limitations. 2024 "actuals" are developed estimates; IBNER assumed 1.0 (±10–30% on long-tail perils); single-year validation folds; motor validation A/E 0.95–0.98; credit band is a fairness-sensitive factor.

2.2 Week 2 — emerging market (gate G6)

What changed. The 2025 test book is 76% new business (train 14%; mean renewal count 1.8 vs 6.9). Adversarial validation separates train from test with AUC 0.86, and `new_business` carries 66–83% of that signal. Against the developed market (2023–24), motor frequency is +35% and severity −45%; contents frequency −15%, severity −10%; buildings is volatile (on-levelled burning cost 569 in 2023, 348 in 2024). Reporting delays and open-claim shares match the developed market. The schema is thinner: no exact age, credit band, mileage or telematics. The data had the same anomaly types as in Week 1, which we cleaned with the same rules (decision #26). Free-text occupation was mapped by rule to the 20 developed levels, with 0 unmapped.

Week 1 model on Week 2 data. The developed model on harmonised features over-predicts emerging 2023–24 costs: A/E motor 0.71, buildings 0.89, contents 0.86, so the **level does not transfer**. The **shape mostly does**: comparing factor relativities between the markets, 31 of 53 motor factors (19/47 buildings, 23/47 contents) can be carried over or have little effect in either market, and only 3 (11, 7) need refitting. Harmonising the features costs motor −0.008 D^2 on developed validation; buildings and contents are almost unchanged.

What was carried over and what changed.

Element	Week 2 choice	Evidence
Rating shape	Motor, contents: harmonised developed blend × emerging level (C1). Buildings: joint developed + emerging fit with a market feature, emerging rows ×5 (C4)	Emerging-only GLMs fail (D^2 motor 0.013, buildings/contents < 0); buildings C4 0.102 vs C1 0.077 (bootstrap band of the gain +0.003 to +0.051)
Level	Re-fitted on emerging 2023–24 (motor ×0.71, contents ×0.86); buildings: credibility Z = 0.5 between the joint fit and emerging actuals	Buildings swings between years; full credibility (≈ 490) and joint fit (≈ 398) bracket the central 446
Development	Developed factors by line × peril	Same reporting delays and open shares
Trend	Developed central (provisional)	Two years are too short to estimate it

Element	Week 2 choice	Evidence
New business	Loading 1.25 on new-business rows (range 1.0–1.5)	See below
Other	Caps and 50/50 blend kept; GBM averaged over 5 seeds (Week 1 lesson). Not used: gender (no lift; fairness), self-employed flag, region factor (rule not met, #34)	Bootstrap rule on train 2023 → validate 2024

New-business loading: the main judgement. Training data are mostly renewals, so the new-business effect is estimated from little evidence. Validation A/E for new business was inconsistent (2024: motor 0.91, buildings 0.57, contents 1.30). We therefore used pre-registered leaderboard probes. Against a loading of 1.0, a loading of 0.9 scored −0.0066 and 1.25 scored +0.0058. To check whether this was a general level effect, we probed a uniform uplift with the same total; it scored 0.0010 below NB 1.25, so we read the effect as specific to new business. We kept 1.25 and did not chase the apparent optimum near 1.28. This choice is **largely leaderboard-informed**: most of the gain (+0.0047 of +0.0058) is shared with a general level increase, so the emerging 2025 level itself may be understated.

Predicted 2025 burning cost (per exposure year): motor 562, buildings 446, contents 99, compared with on-levelled emerging 2023/2024 values of 488/491, 569/348 and 98/93. The motor increase reflects the younger, mostly new-business test mix.

Remaining uncertainty.

Item	Low	Central	High
NB loading	1.0	1.25	1.5
Buildings level (BC)	398	446	490
Trend	Developed low	Developed central	Developed high

The table leaves out four smaller sources of uncertainty:

- We have only two years of emerging history.
- Development factors, including IBNER, are borrowed from the developed market.
- The emerging business may keep changing as the broker book matures.
- The comparison-site commission is blank in `expenses`, although about 5% of policies use that channel (DQ26).

3. Effective use of AI

Workflow. The AI (Claude Code) drafted code, checks and analyses. CLAUDE.md instructed it to act as a challenger, to stop at six human gates (target, data fixes, assumptions, model choice, submission, Week 2 adaptation), and to log every contribution (`logs/ai_usage_log.md`, 17 entries; `logs/decision_log.md`, 38 decisions).

Notable cases.

1. *AI error, caught by documentation:* the AI inferred claims were gross of excess and subtracted it again; the official description said otherwise → corrected before modelling (would have understated cost by ~16.8M a year).
2. *AI omission, caught by validation:* the first LightGBM was not level-balanced (A/E 1.06–1.26); fixed with a training-total balance step.
3. *AI design flaw, caught by the two-fold rule:* prior-claims v1 coded unobservable history (2021) as zero claims — better in 2024, worse in 2023 in every line.
4. *Pre-agreed rule in action:* the AI's recency-weighting idea looked good on 2024 alone (+0.0016) but failed 2023 (−0.0037).
5. *AI probe design (Week 2):* because 76% of test rows are new business, a better score from the NB loading could simply mean the overall level was too low. The AI proposed a uniform-level probe with the same total, which separated the two explanations before we committed to the loading.

6. *AI errors, caught by run checks (Week 2):* the AI had trial-run only one line, so a column missing in emerging motor slipped through. In a second case, an output filter hid a crash. Both were fixed before submission, and every run now records its exit status.

Verification. Every AI-written transform has a unit test (65 tests, incl. leakage property tests); every claim about the data was re-run by the owner on full data; owner-run scripts print compact evidence.

4. Trust, validation & guardrails

- **Validation design:** temporal (train 2021–23 → 2024) plus rolling origin; Week 2 train 2023 → 2024 on emerging with bootstrap bands, and a check on the new-business subset. Acceptance rules were written before results were seen. The leaderboard was used only as a sanity or tie-break check, and each probe was pre-registered with its expected outcome.
- **Diagnostics:** A/E by decile and segment across folds (persistent drifts identified and tested), level sensitivity, per-peril level checks; escalation triggers logged (e.g. motor 2023 A/E 1.13 explained by one claim); Week 2 adversarial validation and PSI between train and test and between markets.
- **Leakage controls:** only information known at inception; prior claims counted only if reported before the row starts. Policy and customer IDs collide across markets, so the pipeline never joins across markets and prefixes IDs in joint fits.
- **Reproducibility:** `python -m src.pipeline --market w1 --model blend` and `python -m src.pipeline_w2`; seeds and assumptions in `configs/config.yaml`. Rebuilds on later days were byte-identical to the submitted files (first submission sha256 72be5f20…, final b3 b1e69ab9…, built by two different scripts). Git tags `w1-final` and `w2-final`.

5. Pricing tool & communication

Pricing tool. A Streamlit tool (`app/pricing_tool.py`) loads the same fitted models as the submission (test-row totals match the final file exactly in every line). For a single policy it returns:

- the expected 2025 cost;
- the GLM and GBM views;
- the experience factor;
- a GLM breakdown (base rate × factor relativities × exposure).

A batch mode prices a CSV and returns it for download, and an About page shows the model, data and assumption versions. Where GLM and GBM disagree for a policy, the team sees both numbers and can check before relying on the blend.

Office premium (Week 2, 20 reference policies). We took our risk premium (RP) and read every rate from `expenses` (`src/office_premium.py`). Loaded cost = RP + cat reinsurance (12% of RP, buildings and contents only) + fixed expense 85 CU. The office premium = loaded cost / (1 − variable 15% − commission by channel − premium tax 8% − profit 8%). For example, B00229821 (buildings, broker):

- RP 325.54
- Cat reinsurance 39.07
- Fixed expense 85.00
- Loaded cost 449.61
- Premium-based loadings 51%
- **Office premium 917.57**, 3.2× its current premium of 285.60

The components add back to the office premium for every policy. Across the 20 policies:

- **The office premium is a median 2.7× the written premium.**
- RP alone is close to the written premium for motor (median 1.02) and buildings (1.20), and well below it for contents (0.48).
- The gap therefore comes mostly from the expenses. Broker business carries 51% premium-based loadings, and the 85 CU fixed expense is close to a typical contents premium (about 100 CU).

- Our reading is that current emerging premiums do not cover expenses and profit, rather than that our risk premium is too high. This is a question about the profitability of the broker book.

Sensitivity of the total.

- Premium tax added on top instead of inside the premium: −6.4%.
- Fixed expense pro-rata to exposure: −2.2%.
- Both: −8.5%.

One outlier, B00217584, has an RP of 1,790 against a written premium of 323. It combines several flood-related features: 0.6 m above river level, 1.7 km from the river, flood zone missing, mixed construction, built in 1933. Current pricing may not reflect the flood exposure or the missing flood zone (DQ27).

6. Limitations & next steps

- **Week 1 lessons carried into Week 2:** average GBMs over seeds by default (applied: +0.0013 median gain on emerging contents); build final candidates early; keep the pre-registered rules.
- **Week 2 lessons:**
- When the test mix differs this much, a single feature (new business) dominates the shift, and two years of history cannot pin down its effect. Monitoring actual-to-expected for the new broker book in 2025 should replace the leaderboard-informed loading as soon as data arrive.
- The emerging trend and IBNER should be re-estimated once a third year is available.
- **Open items:**
- Experience modifier for emerging motor: on emerging validation the gain is +0.0019 (median), with the bootstrap 5th percentile just above zero (+0.0001); new business is treated as history unknown. Not used in the Week 2 finals; to be confirmed on 2025 renewals.
- Per-peril models (frequency learned, severity barely).
- Longer development history for IBNER.
- Fairness review of credit band before any production use (gender was not used in Week 2).
- Commission for the comparison-site channel in the emerging market (DQ26).

Appendix A (not graded). Week 1 leaderboard path

Submission (public / private)	Change vs previous	Public
First blend	—	0.12293
+ long-run weather	+0.0011	0.12401
+ contents GBM c7 (validated)	+0.0004	0.12438
+ buildings c7 + experience modifier (final, private 0.13363)	+0.0004	0.12480

Other probes (within the pre-registered ±0.0005 noise band unless stated): trend low 0.12275, trend high 0.12256, motor loading at 1.19M 0.12307, same-year weather + contents c7 0.12322; after the deadline, 5-seed GBM averaging on b3: public 0.12497 / private 0.13403.

Appendix B (not graded). Week 2 leaderboard path

Submission	Public
C1/C4, buildings credibility 0.5, NB loading 1.0	0.15122
NB 0.9 (probe)	0.14458
NB 1.25 (probe; final candidate 2)	0.15697
Uniform level ×1.2014, same total as NB 1.25 (probe)	0.15596
NB 1.5 (probe)	0.15332
Final: NB 1.25 + 5-seed GBM (final candidate 1)	0.15691